SmartAgent: Multi-agent-based Simulation of Real-world Stock Market Trading

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1 Introduction

The code for this project has been uploaded to a GitHub repository, which can be accessed at: https://github.com/name-available/SmartAgent

1.1 Background

Global Stock Markets represent one of the most complex and dynamic systems in modern finance, characterized by high liquidity, rapid price fluctuations, and interactions among diverse participants including retail investors, institutional funds, and algorithmic trading systems. Market behavior emerges from the interplay of economic indicators (e.g., GDP, interest rates, earnings reports), geopolitical events, and investor sentiment - requiring traders to process vast data streams in real-time to discern meaningful signals from noise.

Traditional approaches rely heavily on historical backtesting (using platforms like Zipline or Backtrader), but face critical limitations: inability to simulate real-time market reactions, susceptibility to overfitting, and neglect of behavioral dynamics among market participants.

The advent of Large Language Models (LLMs) like GPT and Gemini introduces a transformative paradigm for financial simulation. LLM-powered agents can dynamically interpret information and execute trades that approximate real-world decision-making, effectively modeling market responses to unforeseen events (e.g., rate cuts, geopolitical crises).

We present SmartAgent, an innovative LLM-driven multi-agent trading simulation framework comprising three core modules:

- **Pre-Trading Analysis** Agents process macroeconomic data and financial reports to update risk assessments.
- **Trading Execution.** Orders are placed in an event-driven simulated order book.
- **Post-Trading Evaluation.** Performance review and strategy optimization via a Bulletin Board System (BBS)

By integrating LLM reasoning with multi-agent architecture, SmartAgent delivers a more realistic and adaptable platform for quantitative finance research. The following sections detail its design, experimental implementation, and findings - demonstrating how LLM-based simulation can redefine trading strategy development.

1.2 Research objectives

- **Evaluate the Effectiveness of Simulated Trading Behavior** This study aims to assess whether the trading behaviors generated by SmartAgent, when powered by different large language models (LLMs), are both reliable and representative of real-world trading logic. The goal is to determine if the agent's behavior aligns with rational market dynamics.
- **Investigate the Influence of LLMs on Decision-Making** The objective is to examine whether SmartAgent's trading recommendations and strategies are affected by the inherent characteristics or biases of the underlying language models. This includes identifying any systematic tendencies that may influence the agent's outputs.

- **Analyze Simulated Trading Under Varying Market Conditions** This study also seeks to evaluate SmartAgent's ability to autonomously simulate reasonable trading decisions across different market scenarios (e.g., bullish, bearish, or volatile conditions). Furthermore, it investigates how external market factors shape the agent's trading behavior.

2 Experimental Design

Our model design is shown in the figure 1. In our initial experimental design, we used a complete stock trading model based on the real stock market, but it took a long time to implement it; we had to use many tokens, while our budget is limited, so we can only simplify the model. This is why we used a limited number of agents. The design mainly focuses on the modules of our stock day simulation. We divide our smart agent into three main parts: Investment Agent Module, Transaction Module, and BBS Module.

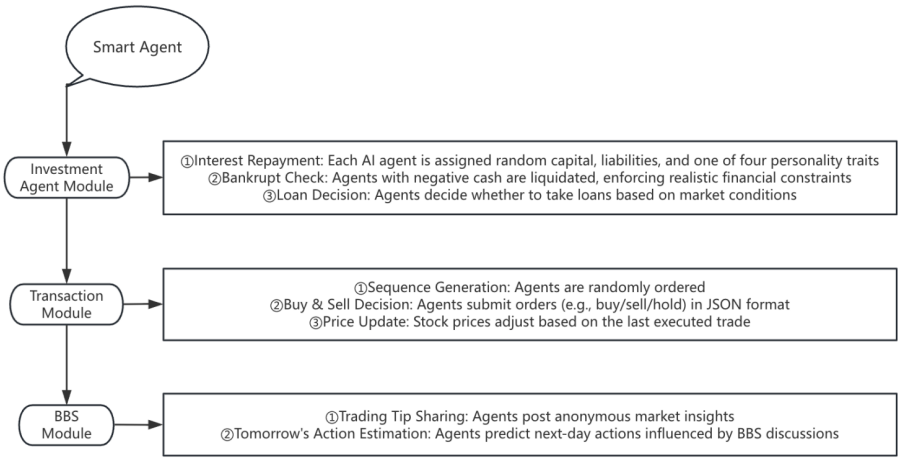

Figure 1: Architecture of SmartAgent model

2.1 Investment Module

Investment Module, is designed for simulating heterogeneous investor behaviors in financial markets. It includes functions like Interest Repayment, Bankrupt Check, Loan Decision and so on. We design it firstly by initializing each AI agent with randomized capital between 100,000 to 5 million units and assign one of four distinct trading personalities: conservative, aggressive, balanced, or growth-oriented. The system incorporates authentic market constraints: variable interest loans from 2.7% to 3.3%, and mandatory bankruptcy procedures when cash turns negative. The breakthrough here is our LLM-powered decision core. Using either GPT or Gemini, agents process real-time market data, financial reports, and BBS discussions to generate JSON-formatted trading decisions.

2.2 Transaction Module

Now to the market engine - Transaction Module. At each session start, we randomize the agent execution sequence using a random clock algorithm, preventing artificial deadlocks that plague traditional simulations. Agents submit orders to a limit order book implemented through optimized dictionary structures. Our trade matching has O(1) complexity. Here we have two key factors: First, we update prices strictly based on last-executed trades. Second, we enforce real-world transaction costs including stamp duties and tiered commissions. The workflow is: random sequencing → order processing → price updates → position reconciliation. This creates what we call "computational market microstructure".

2.3 BBS Module

We have the social dimension, BBS Module. After the market closes, agents post natural language analyses like: Tech sector volatility expected due to supply chain concerns. These anonymous tips are archived and become visible to all agents next trading day, and create an information diffusion network. This directly influences next-day predictions through structured outputs, as agents output decisions combines market data and social sentiment.

3 Experiment Result

3.1 Experimental Setup

The experiment was conducted in a simulated stock market environment comprising two stocks, denoted as Stock A and Stock B. Multiple AI-driven trading agents, powered by Gemini and ChatGPT, were deployed to emulate market traders exhibiting both trading and financial decision-making behaviors. The simulation spanned a period equivalent to one fiscal year, consisting of 264 trading days. Key events, including financial report releases for Stocks A and B, as well as macroeconomic interventions such as reserve requirement ratio cuts and interest rate adjustments, were pre-scheduled to assess agent responses under varying market conditions.

The agents operated in a dynamic environment where their decisions were influenced by real-time market data, including stock prices, financial reports, BBS discussions, and special economic events. As illustrated in the accompanying figure 2, the agents processed these inputs and executed trading actions based on predefined prompt-driven behavioral frameworks.

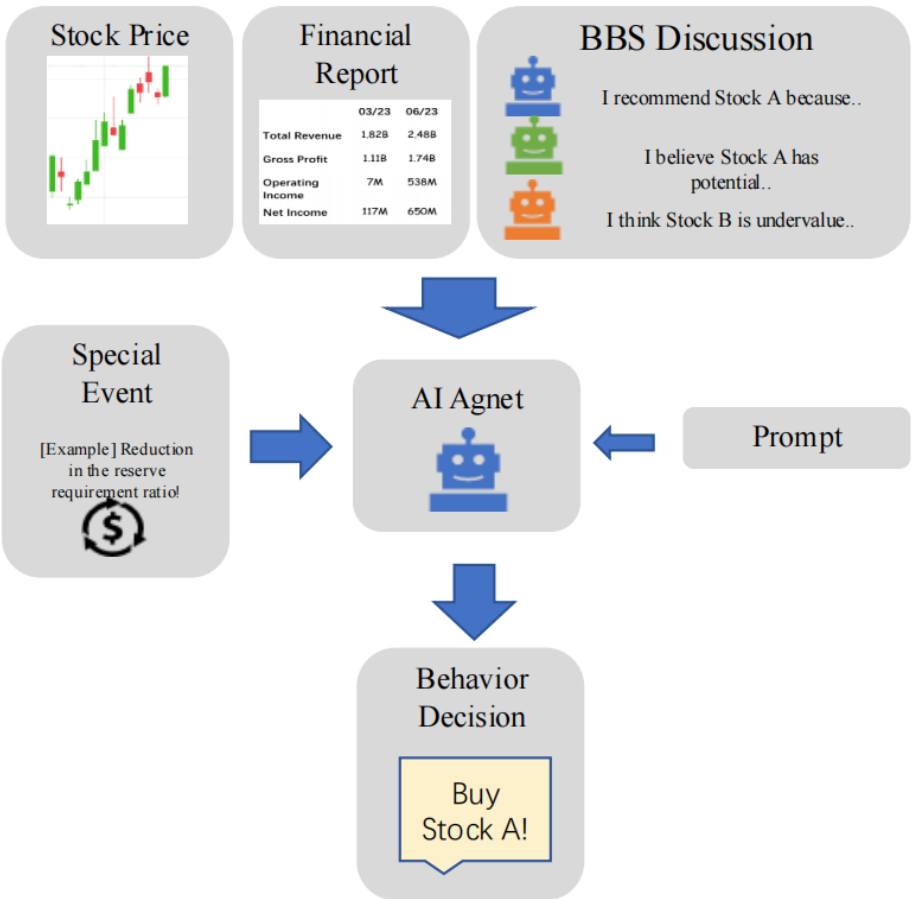

Figure 2: The information sources of AI Agents in making decisions in the simulated market process.

3.2 Price Movement Analysis

The results revealed distinct differences in the trading behaviors of Gemini- and ChatGPT-based agents. Respectively. Notably, the two large language models (LLMs) exhibited divergent trading strategies, prompting further investigation.

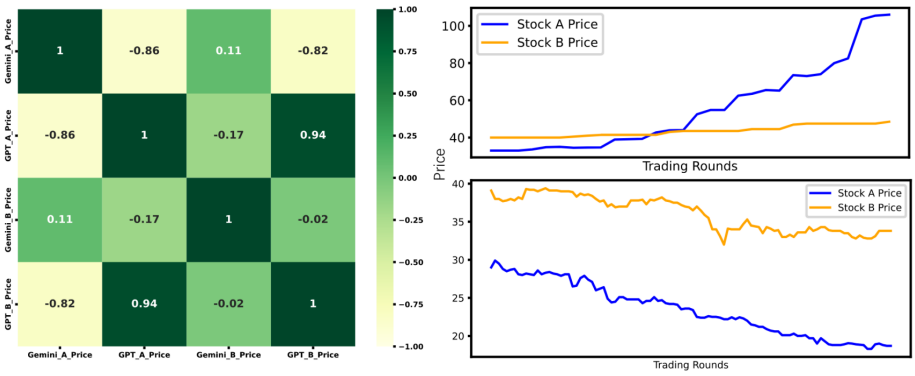

Figure 3: The correlation of price movements of Stock A and B Trading. The left panel of the figure demonstrates the correlation between the price movements of Stocks A and B. The top-right and bottom-right panels depict the stock price trends simulated by ChatGPT and Gemini agents

3.3 Trading Behavior Analysis

A quantitative analysis of trading volumes and frequencies, as presented in the table 4, highlights key behavioral disparities. The ChatGPT agent group demonstrated significantly higher trading volumes but lower trading frequencies compared to the Gemini group. To further elucidate these differences, a K-means clustering algorithm (with k=3) was applied, followed by t-SNE visualization. The clustering results shown in figure 5 indicate that Gemini agents exhibited homogeneous characteristics, with only a minority deviating from the majority trend. In contrast, ChatGPT agents displayed greater heterogeneity in decision-making, engaging in fewer herd- and trend-following transactions, suggesting a higher degree of individual subjectivity.

AI Agent	A Trade Shares	B Trade Shares	A Volume	B Volume
GPT	3118792	329590	176758380	14109526
Gemini	128981	112134	3588331	4325246
AI Agent	**Stock A price**	**Stock B price**	**A Trading Times**	**B Trading Times**
GPT	55.7	43.43	384	263
Gemini	23.46	36.03	800	688

Figure 4: Trading behavior of GPT and Gemini-driven AI Agents.

3.4 Market Sentiment Analysis

The order price change visualization shown in figure 6 reveals distinct sentiment patterns between the two agent groups. Gemini agents tended to adopt a more pessimistic outlook, reflected in their conservative pricing strategies. Conversely, ChatGPT agents exhibited a more optimistic market perspective, leading to comparatively aggressive order placements.

3.5 Detailed Agent Outputs

The experiment logged daily agent activities, including executed trades, portfolio balances, and key decision metrics. Additionally, agents participated in BBS discussions, further enriching the

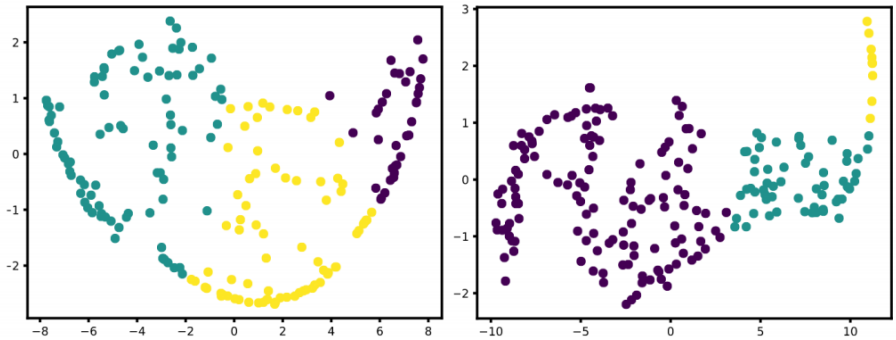

Figure 5: T-SNE visualization of the GPT(Right) and Gemini(Left) agents.

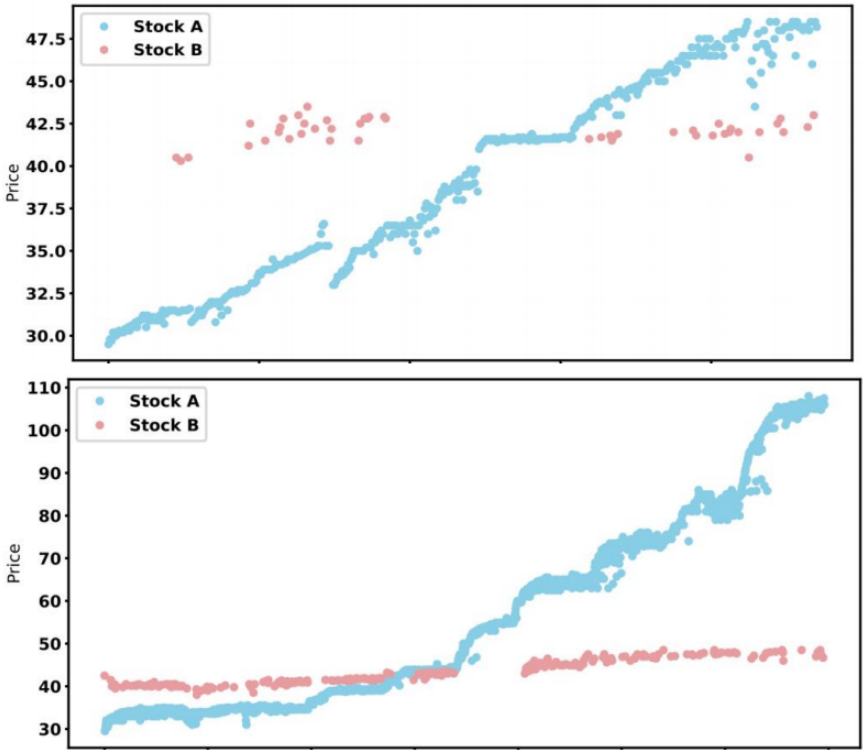

Figure 6: Trading behavior of GPT(Bottom) and Gemini(Top)-driven AI Agents.

simulation with interactive market sentiment dynamics. Figure 7 simply shows the output format. For comprehensive implementation details, please refer to the project homepage https://github.com/name-available/SmartAgent.

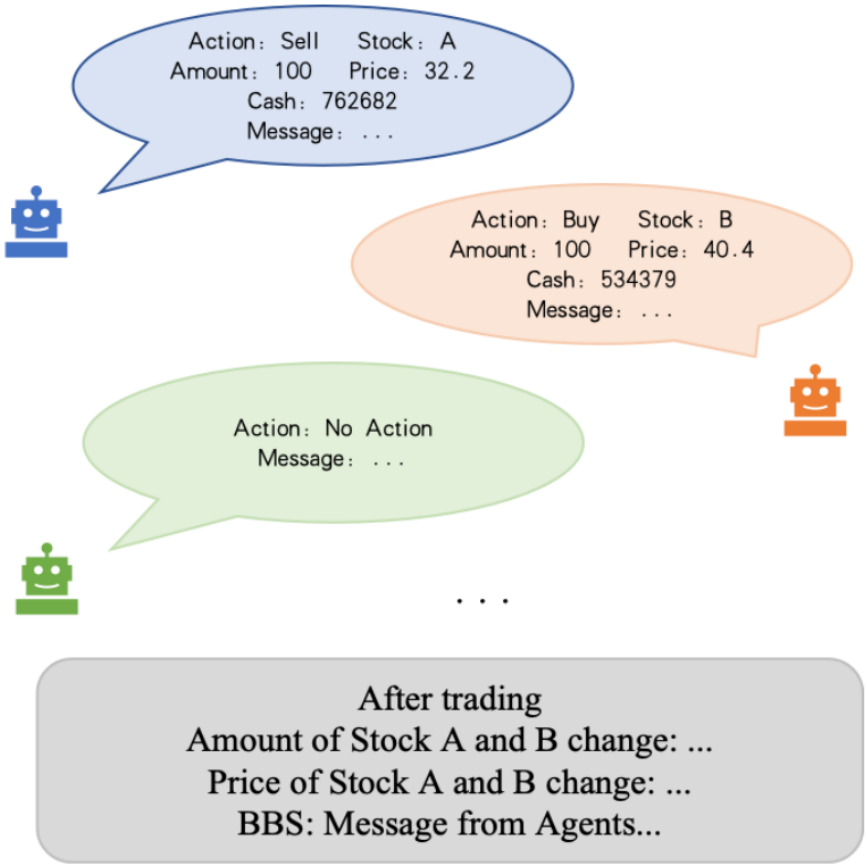

Figure 7: Output format

4 Conclusion

4.1 Key Innovation

- **Different large language models (LLM) lead to differences in trading behavior.**

  AI agent driven by GPT-3.5-Turbo: lower trading frequency but larger trading volume, more cautious towards specific stocks (such as Stock A), individual decision-making differences are significant, and group behavior is diversified (less herd trading).

  AI agent driven by Gemini-Pro: higher trading frequency, similar trading frequency for Stocks A and B, stronger consistency in group behavior (showing "herd behavior").

  The choice of LLM significantly affects the bias of simulation results, users need to pay attention to the differences in market understanding and strategy preferences among models.

- **Heterogeneity of trading behavior within groups.**

  Gemini agent group behavior is more similar, while GPT agent individual differences are large, reflecting more subjective decisions and personalized investment styles.

  Application value: Developers can adjust the role of agent groups based on the needs of the simulation scenario (for example, choose GPT for diversity and Gemini for consistency).

- **External factors affecting trading behavior.**

  Disabling non-first-round loans and BBS features: trading strategies of agents become more conservative.

  Removing profit reports and interest rate changes: some agents turn from losses to profits.

  Removing financial information support: exacerbates the differences in gains and losses among agents, making market competition more intense.

4.2 Future Work

- **Parameter Influence.** Explore the impact of different parameters (such as different numbers of AI agents, more stocks, different trading days, different BBS sharing times) on the simulation results.
- **Ablation experiment** Explore the impact of the missing BBS module, Financial information, etc. on the simulation results.
- **Applying Deepseek.** Due to time and budget constraints, this project did not use deepseek for simulation. The simulation effect of deepseek will be supplemented later.

4.3 Conclusion

SmartAgent is an AI agent framework driven by LLMs aimed at simulating stock trading in real-world environments. SmartAgent replicates the processes and external conditions of actual trading, offering valuable insights into AI Agent-based stock trading methods. The framework's design purpose is to alleviate the impact of historical stock trends on the prior knowledge of LLMs, allowing AI agents to explore diverse trading patterns and preferences in stock trading environments.